BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: Mathematics Paper–X (02) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** Mathematics Paper–X (02) (23)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. For the quadratic equation $ax^2 + bx + c = 0$, If $b^2 - 4ac = 0$ then roots are.....

(A) Unequal
(B) Imaginary
(C) Real
(D) Equal

Q2. If A and B are disjoint sets then....

(A) $A \cap B = \phi$
(B) $A \cup B = \phi$
(C) $A \cap B = A$
(D) $A \cup B = A$

Q3. A chord is 5 cm from the center of circle of radius 13 cm. The length of the chord is.....

(A) 6 cm
(B) 12 cm
(C) 24 cm
(D) 30 cm

Q4. In a circle two chords are equidistant from the center of a circle, the chords are.....

(A) Congruent
(B) Not congruent
(C) Parallel
(D) Non-parallel

Q5. How many tangents can be drawn from a point outside the circle?

(A) 1
(B) 2
(C) 3
(D) 4

Q6. If "f" is a function from A to B, then "f" is onto function if:

(A) Range f ≠ A
(B) Dom f ≠ A
(C) Range f = B
(D) Range f ≠ B

Q7. The formula Σx/n determines.....

(A) Arithmetic mean
(B) Median
(C) Mode
(D) Geometric mean

Q8. $1 + \tan^2 \theta$ =.....

(A) $\sin^2 \theta$
(B) $\cos^2 \theta$
(C) $\text{cosec}^2 \theta$
(D) $\sec^2 \theta$

Q9. The most frequent value in the data is called its.....

(A) Mean
(B) Median
(C) Mode
(D) Geometric mean

Q10. 45° = ...radians.

(A) π/3
(B) π/4
(C) π/6
(D) π/2

Q11. If $x^2 - 3x - 4 = 0$, then the solution are.....

(A) x = 4, –1
(B) x = –4, 1
(C) x = 4, 1
(D) x = –4, –1

Q12. Which of the given values of x are the roots of equation $x^2 - 8x + 15 = 0$

(A) x = 1 or x = 7
(B) x = 2 or x = 4
(C) x = –2 or x = 4
(D) x = 3 or x = 5

Q13. If $m \propto 1/n$, then:

(A) m = kn
(B) n = km
(C) m/n = k
(D) mn = k

Q14. If 2a + 1 : 21 :: 4 : 7, then

(A) a = 13/2
(B) a = 11/2
(C) a = 10
(D) a = 9/2

Q15. If P(x) and Q(x) are two polynomials then P(x)/Q(x), Q(x) ≠ 0 is:

(A) Rational fraction
(B) Irrational fraction
(C) Proper fraction
(D) Improper fraction